Bipanshu Kashyap

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CAREER OBJECTIVE

B.Tech Computer Science student at Graphic Era University (GPA 9.00 / 10.00) with AWS-certified expertise in machine learning pipelines and generative AI. Proficient in Python, C++, Flask, TensorFlow, and Node.js with hands-on experience building AI-powered, full-stack, and algorithm-driven applications. Seeking a software engineering or machine learning internship to contribute to production-scale systems. Scored **98.00%** in Class X and completed **2 AWS certifications** within the first year of undergraduate study.

EDUCATION

Graphic Era University — **Aug 2024 – Aug 2028 (Expected)**
B.Tech Computer Science — GPA: ***9.00 / 10.00*** — *Dehradun, Uttarakhand*

Vidhya Bharti Senior Secondary School — **2023**
Senior Secondary — R.B.S.E (Class XII) — ***89.40%*** — *Rajasthan*

Adarsh Vidhya Mandir Secondary School — **2021**
Secondary — R.B.S.E (Class X) — ***98.00%*** — *Rajasthan*

EXPERIENCE

Freelance Web Developer — **May 2026 – Sep 2026**
Self-Employed — HTML, CSS, JavaScript, Responsive Web Design — *Remote*

- Designed, built, and deployed **2 live client websites** end-to-end — Sheeraj Events (a luxury wedding and event-planning business) and Shaitan Singh Vavdi (a professional stage anchor's personal brand site) — owning requirements gathering, UI design, development, and deployment.
- Developed responsive, mobile-first interfaces featuring service showcases, image galleries, and inquiry/contact forms, with integrated WhatsApp and social-media links to drive client bookings.
- Worked directly with business owners to translate their goals into clean, on-brand designs, iterating on feedback through to production launch.

PROJECTS

ADOC — Write-Stall Mitigation for RocksDB — **Oct 2026 – Present**
Developer — C++17, RocksDB, CMake, AIMD Control, LSM-Tree Storage — *Personal Project*

- Reproduced the **ADOC** control algorithm (USENIX FAST '23) in C++ to suppress write stalls in RocksDB, polling the engine **once per second**, classifying the overflowing component (MemTable, Level-0, or pending-compaction redundancy), and auto-retuning **2 knobs** (background thread count and batch size) via an **AIMD** policy inspired by TCP congestion control.
- Integrated the tuner with a live RocksDB through a custom `rocksdb::EventListener`; on a **60-second, 4-writer** benchmark (RocksDB 9.11) it cut write-stall time from **15.63 s to 0 s** while sustaining **138 kOps/s** throughput.
- Built a synthetic LSM dataflow simulator sharing the same tuner and a **23-case conformance suite** (all passing) validating overflow detection, AIMD step sizes, priority ordering, and knob bounds against the paper.

Emotion Detection System — **Feb 2026 – May 2026**
Developer — Python, TensorFlow, Keras, OpenCV, Deep Learning — *Personal Project*

- Developed a real-time facial emotion recognition system using Convolutional Neural Networks trained on the FER-2013 dataset (**35,887 images**) across **7 emotion classes**, achieving **90%+ validation accuracy** after 50 training epochs.
- Integrated OpenCV for live webcam face detection and emotion classification, delivering prediction latency of **under 200 ms per frame** for smooth real-time performance.
- Applied image preprocessing (grayscale normalization, face cropping), data augmentation (rotation, flipping, zoom), and dropout regularization, improving model robustness across varying lighting and facial orientations.
- Designed a **5-layer CNN architecture** with batch normalization optimized using the Adam optimizer, reducing training loss by **35%** over baseline across the full training schedule.

Friend Recommendation System with Token Economy **Jan 2026 – Feb 2026**
Developer — C++, Node.js, Express.js, SQLite, JavaScript *Graphic Era University — DAA Project*

- Engineered a full-stack social networking app with a graph-based Top-K recommendation engine (C++) scoring user similarity across shared interests and location, returning the **top 3 matches** per query from a live user pool.
- Designed a token economy — users earn tokens on daily login, spend **2 tokens** per friend request and **1 token** per recommendation view, and receive mutual rewards on accepted connections.
- Built a Node.js + Express REST API with **5 endpoints** (login, register, friend requests, recommendations, chat) backed by SQLite, achieving response times **under 100 ms** per request.
- Implemented real-time chat using WebSocket connections and a responsive vanilla JS frontend, enabling live messaging between connected users without page refresh.

GameHub — Centralized Gaming Platform **Aug 2025 – Dec 2025**
Developer — Python, SQLite, Object-Oriented Programming *Graphic Era University*

- Consolidated **4 games** under a single authenticated interface, eliminating the need for separate standalone executables and reducing user launch friction.
- Built a SQLite-backed scoreboard tracking **3 per-game statistics** (score, global rank, cumulative play count) with persistent cross-session user history.
- Adopted a plugin-style modular architecture reducing new-game integration to **under 30 minutes**, requiring only **1 configuration file** per game module.
- Implemented role-based access control separating guest and registered user permissions, improving data isolation and overall platform security.

Train Reservation System **Sep 2024 – Nov 2024**
Developer — C++, Python, Data Structures and Algorithms *Personal Project*

- Automated **3 core operations** (booking, viewing, cancellation) across **10 train routes** with up to **100 seats per train** through a menu-driven console application.
- Designed a **3-file persistent storage architecture** (passengers, seats, bookings) maintaining data integrity for **500 records** with zero data loss under stress testing.
- Applied array-based seat allocation with **O(n) in-place update** strategy, achieving consistent booking operation performance across full system load.
- Built comprehensive input validation and error handling for all user operations, preventing invalid bookings and ensuring data consistency across concurrent session interactions.

TECHNICAL SKILLS

- **Programming Languages:** Python (Proficient), C++ (Proficient), C (Proficient), JavaScript
- **Cloud and AI Platforms:** Amazon Web Services (AWS SageMaker, AWS ML Pipeline, AWS Generative AI), Gemini API, Claude Code
- **Machine Learning and AI:** TensorFlow, Keras, PyTorch (Learning), scikit-learn, NumPy, Pandas, Natural Language Processing, Supervised Learning, Generative AI, Deep Neural Networks, Convolutional Neural Networks, OpenCV
- **Concepts:** Data Structures and Algorithms, Graph Algorithms, Probability and Statistics, Linear Algebra, Feature Engineering, Model Evaluation, Cross-Validation, Object-Oriented Programming, REST APIs, Cloud Computing
- **Frameworks and Tools:** Flask, Node.js, Express.js, SQLite, Git, GitHub, VS Code, MATLAB

CERTIFICATIONS

The Machine Learning Pipeline on AWS — AWS Training & Certification *Oct 2025*
Developing Generative Artificial Intelligence Solutions — AWS Training & Certification *Dec 2025*